

Comparison of Imaging Findings between Granulomatosis with Polyangiitis and Eosinophilic Granulomatosis with Polyangiitis on Sinus CT: Importance of High-Density Opacification of the Paranasal Sinuses

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ABSTRACT

BACKGROUND AND PURPOSE: Granulomatosis with polyangiitis (GPA) and eosinophilic granulomatosis with polyangiitis (EGPA) are the most common causes of chronic sinusitis from systemic granulomatous diseases. While both are small- to medium-sized vasculitis with necrotizing granulomas, they have different clinical courses and prognoses. High-density sinus opacification has been reported in allergic fungal sinusitis with eosinophilic infiltrates. Given that EGPA also has eosinophilic tissue infiltrates, we evaluated the differences in sinus CT findings, focusing on the sinus secretion attenuation between patients with GPA and EGPA, along with other previously described findings.

MATERIALS AND METHODS: This study included 31 patients with GPA and 22 patients with EGPA who underwent sinus CT. The attenuation of secretions within the paranasal sinuses was visually assessed, and the Hounsfield unit (HU) of the highest-density portions within each sinus was measured. Lund-Mackay scores (LMS), bony destruction, sclerotic wall changes, adjacent organ involvement, and nasal polyps were evaluated and compared between patients with GPA and EGPA. Multiple logistic regression analyses were conducted to determine which factors independently discriminated GPA from EGPA, and the diagnostic ability to differentiate between these 2 diseases was evaluated by using a receiver operating characteristic curve analysis.

RESULTS: More patients in the GPA group showed bony destructions, bone sclerosis, and involvement of organs adjacent to paranasal sinuses than in the EGPA group ($P = .006$, 0.048 , and 0.035 , respectively). The EGPA group had higher LMS and more nasal polyps than the GPA group ($P = .078$ and 0.333 , respectively). More patients in the EGPA group showed internal high-density opacification than in the GPA group, and patients with EGPA had higher mean HUs (both $P < .0001$). The presence of high-density opacification or mean HUs independently distinguished GPA from EGPA (OR, 53.67 and 1.07 ; 95% CI, 4.07 – 708.03 and 1.02 – 1.13 , respectively) and showed a greater ability to discriminate between these diseases compared with other findings.

CONCLUSIONS: Patients with EGPA had more high-density sinus opacification and higher mean HU on sinus CT than the patients with GPA. In addition to the previously reported CT findings, such as bony destruction, bone sclerosis, and adjacent organ involvement, evaluating secretion attenuation can assist in distinguishing between GPA and EGPA.

ABBREVIATIONS: AUROC = area under the receiver operating characteristic curve; EGPA = eosinophilic granulomatosis with polyangiitis; GPA = granulomatosis with polyangiitis; HU = Hounsfield unit; LMS = Lund-Mackay score; ROC = receiver operating characteristic

Chronic rhinosinusitis is a prevalent disease affecting 7%–14% of people in North America.^{1,2} It commonly presents with chronic nasal obstruction, rhinorrhea, and nasal crusting. Even

though many of the cases have an infectious or allergic etiology, these symptoms can be from systemic granulomatous diseases such as granulomatosis with polyangiitis (GPA, Wegener granulomatosis) or eosinophilic granulomatosis with polyangiitis (EGPA, Churg-Strauss syndrome). While both are small- to medium-sized vasculitis with necrotizing granulomas, they have different clinical courses and prognoses. Because paranasal sinusitis is an early manifestation of these diseases and may precede other organ involvement, it is important to differentiate and diagnose these diseases as early as possible based on sinus imaging characteristics.³

The prevalence of GPA in the United States is estimated to be 3 cases per 100,000 people.⁴ GPA involves the head and neck, lungs, kidneys, and central nervous system. Approximately 85% of

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SUMMARY

PREVIOUS LITERATURE: Granulomatosis with polyangiitis (GPA) and eosinophilic granulomatosis with polyangiitis (EGPA) are the most common causes of chronic sinusitis from systemic granulomatous diseases. Both are small- to medium-sized vasculitis with necrotizing granulomas. While both diseases frequently appear as a nonspecific chronic sinusitis pattern in the early stages, they have different clinical courses and prognoses. Previous articles on allergic fungal sinusitis have shown that mucin with high eosinophilic infiltration in the paranasal sinuses has high attenuation on CT. Because EGPA also involves hypereosinophilic tissue infiltrates, the analysis of intrasinus attenuation on CT can help to differentiate between the 2 diseases.

KEY FINDINGS: The patients with EGPA had more high-density sinus opacification and higher mean HUs on sinus CT than the patients with GPA. The presence of high-density sinus opacification or mean HUs independently distinguished GPA from EGPA and showed higher diagnostic ability on CT to discriminate between these diseases compared with other CT findings.

KNOWLEDGE ADVANCEMENT: In addition to the previously reported CT findings, such as bone destruction, bone sclerosis, and adjacent organ involvement, evaluating secretion attenuation on sinus CT can assist in distinguishing between GPA and EGPA.

patients with GPA demonstrate head and neck involvement, with the sinonasal region being the most frequently involved.^{3,5} Isolated paranasal and sinonasal involvement is present in 25% of cases. GPA frequently manifests as a nonspecific chronic sinusitis pattern, especially early in the disease course.^{3,5}

The incidence of EGPA in the United States is 1–3 cases per 100,000 people per year.⁶ EGPA is characterized by marked eosinophilia and necrotizing granulomas with internal eosinophilic infiltrates. The nasal cavity and the paranasal sinuses are the most frequently affected regions.^{3,7} Chronic rhinosinusitis is the most common initial symptom and is present in up to 48%–86% of EGPA cases.^{3,7,8} In the prodromal stage, it shows rhinosinusitis and progresses to hypereosinophilia and eosinophilic tissue infiltrates. This is followed by the development of systemic vasculitis symptoms and multiorgan involvement, including the lungs, nerves, vascular systems, and kidneys.^{3,7} It was reported that approximately 12% of patients with EGPA died within a year without adequate diagnosis and treatment.⁹

In addition to the clinical, serologic and pathologic findings, imaging findings play an important role in developing an early differential diagnosis, defining the extent of the disease, and monitoring/surveillance disease activity during and after treatment of chronic granulomatous sinusitis. CT is the primary technique for diagnosing and assessing inflammatory diseases of the paranasal sinuses.¹⁰ Previous articles on allergic fungal sinusitis have shown that mucin with high eosinophilic infiltration in the paranasal sinuses has high CT attenuation.^{11–14} Because EGPA also involves hypereosinophilic tissue infiltrates, this study aims to compare the CT imaging findings between GPA and EGPA, focusing on the internal attenuation of the sinuses as well as other well-known CT findings.

MATERIALS AND METHODS

This retrospective study was performed across 2 large academic medical centers (Massachusetts General Hospital and Boston Medical Center). Institutional review board approval was obtained at each hospital, and a waiver of informed consent and Health Insurance Portability and Accountability Act authorization were obtained at each hospital. The methodology proposed in the Standards for Reporting of Diagnostic Accuracy Studies checklist was followed.

Patients

We reviewed all patients diagnosed with GPA or EGPA who underwent a noncontrast sinus CT between January 2001 and December 2023 at these 2 tertiary hospitals. Fifty-three patients with GPA and 32 patients with EGPA were identified. Among them, 22 patients with GPA and 10 with EGPA were excluded due to the lack of discernible sinusitis on CT or poor diagnostic image quality that hindered evaluation. A total of 31 patients with GPA and 22 patients with EGPA were included in this study: 17 patients with GPA and 12 patients with EGPA from 1 hospital, and 14 patients with GPA and 10 patients with EGPA from another hospital. The diagnosis of GPA and EGPA was made based on histologic diagnosis from biopsy specimens obtained from the kidney, nasal cavity, paranasal sinus, or lung in 9 patients with GPA and 8 patients with EGPA. Other patients were diagnosed based on the clinical criteria for each disease.

CT Imaging

All patients underwent noncontrast paranasal sinus CT scans with 16- to 64-detector row CT (IQon Spectral CT [Philips Healthcare]; LightSpeed VCT, Bright Speed, Discovery CT 750 HD, Revolution CT, Optima CT660, HiSpeed CT [GE Healthcare]; Somatom force, Biograph 64 [Siemens Healthineers]) by using a tube voltage of 90–140 kVp and a tube current of 160–275 mA. CT images, with 1.0- to 2.0-mm slice thickness, were reconstructed into axial, coronal, and sagittal images with soft tissue and bone algorithms by using iterative reconstruction algorithms on each vendor's scanner.

Imaging Analysis

Evaluation of High-Density Sinus Secretions. The attenuation of sinus secretions was visually assessed for the presence or absence of high-density material within the paranasal sinuses. This was defined as sinus secretions having a visually apparent or subjective high-density component compared with the adjacent mucosa or submucosa. The Hounsfield units (HUs) of the subjective high-density portion were also measured, and the HUs of the highest-density portion in the paranasal sinuses of all patients were measured.

Table 1: Characteristics of patients with GPA and EGPA

Group	GPA (n = 31)	EGPA (n = 22)
Age mean SD (years)	50.1 ± 20	50.8 ± 13.6
Sex (M:F)	15:16	13:9
ANCA status		
c-ANCA	20	0
p-ANCA	2	6
Both c-, p-ANCA	1	0
Negative	8	13
N/A	0	3
Eosinophil		
Eosinophilia (>8%)	1	20
Normal	30	0
N/A	0	2

Note:—SD indicates standard deviation; ANCA, antineutrophil cytoplasmic antibody; N/A, not available.

Evaluation of Other CT Findings. The presence of any bony destruction of the nasal cavities and paranasal sinuses (including the nasal septum), sclerotic changes of the paranasal sinus walls, adjacent organ involvement including the orbital cavity, nasolacrimal ducts, and intracranial region, and the presence of low-density nasal polyps were also assessed and recorded by using a binary scoring system (positive versus negative).

The CT images were also evaluated by using the Lund-Mackay score (LMS), which evaluates the extent of sinusitis in 10 regions derived from the left and right paranasal sinuses (maxillary, sphenoidal, anterior ethmoidal, posterior ethmoidal, and frontal) and assigned a grade of 0–2 based on the degree of involvement. The LMS also evaluates whether the ostiomeatal complex is obstructed on each side, and a score of 0 or 2 is assigned based on the extent of involvement. Twelve regions were evaluated for a total score of 24 points.^{15,16}

All images were evaluated independently by 2 head and neck radiologists with more than 10 and 25 years of experience, and the discrepancy was resolved by consensus.

Statistical Analysis

All statistical analyses were performed by using MedCalc software for Windows (MedCalc software V, 22.0.2.1). A 2-tailed *P* value ≤ .05 was considered significant for all statistical analyses. The HUs of the portion from the highest attenuation secretions in the paranasal sinuses were compared between the GPA and EGPA groups by using the unpaired Student *t* test. The presence of subjective high-density secretions, bone destruction, sclerotic change of the sinus walls, adjacent organ involvement, and nasal polyps was compared between the 2 groups by using the χ^2 test or Fisher exact test. The LMS was compared between the 2 groups by using the Mann-Whitney *U* test. The distribution of CT energy levels between the 2 groups was also compared by using Fisher exact test.

Multiple logistic regression analyses were conducted to determine which factors independently discriminated GPA from EGPA among the factors with a *P* value < .1 in a univariate analysis. First, it was conducted with the presence of subjective high-density lesions and then with the HU. All factors with a *P* value < .05 were considered independent factors to discriminate between the 2 diseases.

The diagnostic ability to differentiate between the 2 diseases was evaluated by generating a receiver operating characteristic (ROC) curve. The optimal cutoff values for differentiating between the 2 groups were identified by maximizing the Youden index.

RESULTS

Patient Characteristics

There were 15 men and 16 women in the GPA group and 13 men and 9 women in the EGPA group. The mean age of the GPA and EGPA groups was 50.1 ± 20 years (range, 19–88) and 50.8 ± 13.6 years (range, 25–79), respectively. No significant difference in patient age was noted between the GPA and EGPA groups (*P* = .889). Table 1 summarizes patient data on antineutrophil cytoplasmic antibody and eosinophil status. In the GPA group, 1 patient was scanned with 90 kVp, 4 patients with 140 kVp, and 26 patients with 120 kVp. In the EGPA group, 3 patients were scanned with 140 kVp and 19 patients with 120 kVp. There was no significant difference in the distribution of CT energy levels between the 2 groups (*P* = 1.0).

Analysis of High-Density Paranasal Sinus Secretions

Sixteen of 22 patients (73%) with EGPA showed subjective high-density material in the paranasal sinuses compared with 5 of 31 patients (16%) with GPA who had high-density material in the paranasal sinuses (*P* < .0001) (Figs 1, 2A, -B). The mean HUs of the subjective high-density components of the patients with EGPA and that of the patients with GPA were 94.1 ± 18.7 (range, 66.4–123.1) and 81.4 ± 20.8 (range, 62.9–114.5), respectively, which showed no significant difference between the groups (*P* = .674).

In the analysis of the HUs corresponding to the highest-density secretions in the paranasal sinuses of all patients, the mean HU of the patients with EGPA (77.8 ± 31.9; range, 24.1–123.1) was significantly higher than that of the patients with GPA (44.4 ± 20.2; range, 19.9–114.5) (*P* < .0001). The ranges of the HUs from the patients with EGPA and GPA who did not have subjective high-density components in the paranasal sinuses were 24.1–45.6 and 19.9–50.6, respectively.

Analysis of Other CT Findings

Fourteen of 31 (45%) patients with GPA had bone destructions, including nasal septa, sinus walls, orbital walls, and skull base (Fig 3). Two patients of the EGPA group (9%) showed destruction of the nasal septum (*P* = .006). The GPA group showed more sclerotic changes of the paranasal sinus walls (*n* = 16) (Figs 2C, -D, 3) than the EGPA group (*n* = 5) (*P* = .048). Six patients in the GPA group showed adjacent organ involvement (Fig 3), whereas no patients with EGPA showed adjacent organ involvement (*P* = .035). Nasal polyps were more prevalent in the EGPA group (10 of 22 patients) compared with the GPA group (10 of 31 patients) (Figs 1, 2A, -B), however, statistical significance was not achieved (*P* = .333).

The mean LMS of the EGPA group (13.1 ± 6.3; range, 4–21) was higher than that of the GPA group (10.1 ± 6.5; range, 2–24), however, statistical significance was not achieved (*P* = .078). All the paranasal sinus CT imaging findings of the patients with GPA and EGPA are summarized in Table 2.

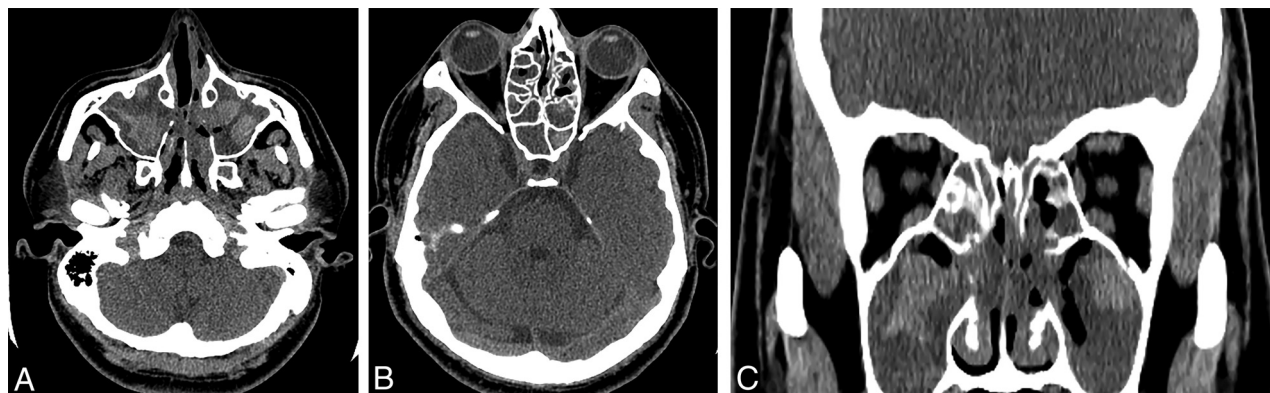


FIG 1. A 28-year-old woman with eosinophilic granulomatosis with polyangiitis. *A and B*, Axial paranasal sinus CT images show high-density sinus opacifications in both maxillary and left sphenoidal sinuses and low-density nasal polyps in both nasal cavities. HU of high-density materials in the right maxillary, left maxillary, and left sphenoid sinuses are 66.7, 70.0, and 69.2, respectively. *C*, Coronal paranasal sinus CT image shows high-density materials in both maxillary sinuses and low-density nasal polyps in both nasal cavities.

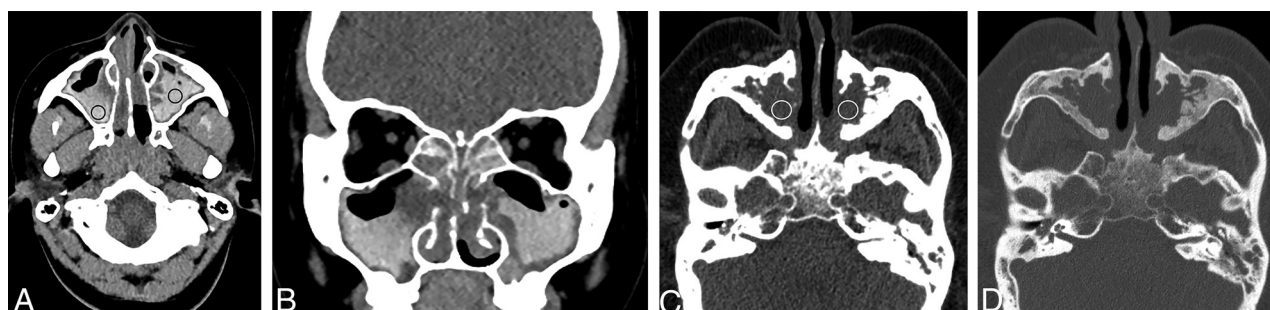


FIG 2. *A and B*, A 25-year-old woman with eosinophilic granulomatosis with polyangiitis. *C and D*, A 48-year-old man with granulomatosis with polyangiitis. *A*, Axial paranasal sinus CT image shows high-density sinus opacifications in both maxillary sinuses and low-density nasal polyps in both nasal cavities. HU of high-density materials in the right and left maxillary sinuses are 91.6 and 91.5 (black circular ROIs), respectively. *B*, Coronal paranasal sinus CT images show high-density materials in both maxillary and ethmoidal sinuses and low-density nasal polyps in both nasal cavities. *C*, Axial paranasal sinus CT image with soft tissue algorithm shows internal low-density materials in both maxillary sinuses. HU of low-density materials in the right and left maxillary sinuses are 43.5 and 41.5 (white circular ROIs), respectively. *D*, Axial paranasal sinus CT image with bone algorithm shows destruction of the medial walls and diffuse sclerosis of the anterior and posterior walls of maxillary sinuses bilaterally.

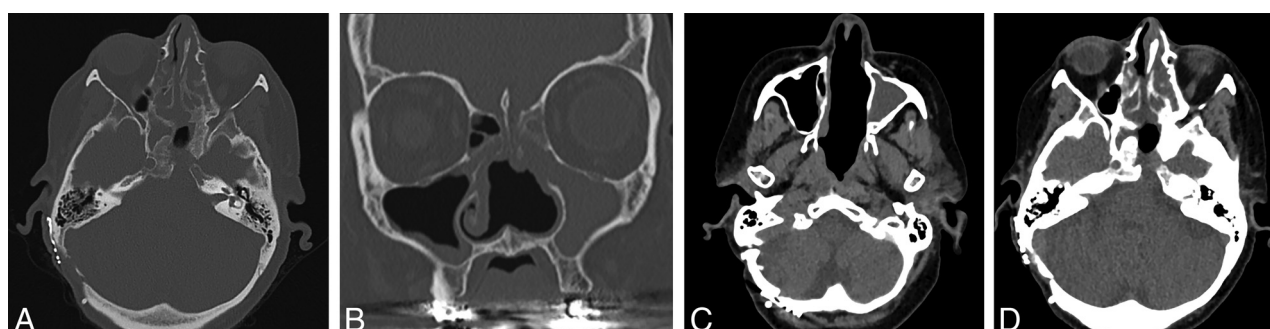


FIG 3. A 53-year-old woman with granulomatosis with polyangiitis. Axial (*A*) and coronal (*B*) paranasal sinus CT images with bone algorithm show diffuse sclerotic wall thickening of left sphenoidal sinus wall with posterior wall destruction (*A*) and bone destructions in nasal septum and nasal turbinates (*B*). Axial paranasal sinus CT images with soft tissue algorithm show left maxillary sinusitis without internal high-density (*C*) and suspicious intracranial extension of soft tissue attenuation lesion in the left sphenoidal sinus (*D*).

Multiple Logistic Regression Analysis

Multiple logistic regression analysis, including factors with a P value $< .1$ in the univariate analysis, showed that only the presence of subjective high-density sinus secretions ($P = .003$) or the mean HU of the highest-density portion of the secretions

($P = .005$) was able to differentiate GPA from EGPA on the paranasal sinus CT. The presence of bone destruction, sclerotic bone change, adjacent organ involvement, and LMS were not independent factors. The results of multiple logistic regression analyses, one with the mean HU of the highest-density portion

and another with the presence of subjective high density, are summarized in Table 3 and 4.

Diagnostic Ability Analysis

In the ROC curve analysis of the mean HUs from each group, the area under the receiver operating characteristic curves (AUROCs) for differentiating GPA from EGPA was 0.777 (95% CI, 0.642–0.88). The threshold discriminating between GPA and EGPA was 64.3. The AUROC for the presence of subjective high-density sinus secretions differentiating between the 2 diseases was 0.783 (95% CI, 0.648–0.884).

For other CT image findings, AUROCs for the presence of bone destruction, sclerotic bone change, and adjacent organ involvement for discriminating GPA from EGPA were 0.68, 0.597, and 0.644, respectively. All the AUROCs of imaging

findings in differentiation between GPA and EGPA with 95% CI, sensitivity, and specificity were summarized in Table 5.

DISCUSSION

Many systemic granulomatous diseases, such as GPA, EGPA, and sarcoidosis, may manifest as chronic rhinosinusitis. GPA and EGPA are the most common of these entities. Both diseases are systemic small-to-medium-sized vessel vasculitis with necrotizing granulomas, with rhinosinusitis being the most common presenting manifestation.^{3,17-19} However, as the name suggests, unlike GPA, EGPA lesions usually have eosinophilic tissue infiltrates along with hypereosinophilia.^{3,15,20}

Previous literature on allergic fungal sinusitis describes high-density opacifications in the paranasal sinus CT. Histologic examinations of these patients demonstrate sinus material containing allergic mucin composed of mucin, eosinophils, and Charcot Leyden crystals without fungal materials. This allergic mucin shows high attenuation on CT and low signal intensity on both T1- and T2-weighted MR images due to its high protein concentrations (more than 28%) and low free water.¹¹⁻¹⁴

In this study, the patients with EGPA showed higher mean HUs in the paranasal sinuses than the patients with GPA. In addition, more patients (73%) in the EGPA group showed subjective high-density secretions than in the GPA group. All the patients with subjective high-density secretions had HUs higher than 60 HU (the lowest value

Table 2: The paranasal sinus CT findings of patients with GPA and EGPA

	GPA (n = 31)	EGPA (n = 22)	P
HU (mean ± SD)	44.4 ± 20.2	77.8 ± 31.9	< .0001
High-density component (present:absent)	5:26	16:6	< .0001
Bone destruction			.006
Nasal septum	7	2	
Nasal septum + sinus wall	4	0	
Nasal septum + sinus wall + orbital wall	2	0	
Sinus wall + orbital wall + skull base	1	0	
None	17	20	
Sclerotic change (present:absent)	16:15	5:17	.048
Organ involvement			.035
Orbit	3	0	
Nasolacrimal duct	1	0	
Orbit + nasolacrimal duct	1	0	
Intracranial extension	1	0	
None	25	22	
Nasal polyp (present: absent)	10: 21	10: 12	.333
LMS (mean ± SD)	10.1 ± 6.5	13.1 ± 6.3	.078

Table 3: Multiple logistic regression analysis with the presence of subjective high-density in the paranasal sinuses

Imaging Finding	Coefficient (B)	Standard Error	P	OR	95% CI
High-density	3.983	1.316	.003	53.666	4.068–708.032
Bone destruction	−2.481	1.35	.066	0.084	0.006–1.18
Sclerotic change	−0.633	0.925	.494	1.884	0.307–11.552
Organ involvement	−19.417	6640.508	.998	3.69E-09	0-Exp (12,995)
LMS	−0.071	0.085	.401	0.932	0.789–1.099

Table 4: Multiple logistic regression analysis with the mean HU of the highest-density portion in the paranasal sinuses

Imaging Finding	Coefficient (B)	Standard Error	P	OR	95% CI
Mean HU	0.067	0.025	.005	1.072	1.021–1.125
Bone destruction	−1.842	1.224	.132	0.159	0.014–1.746
Sclerotic change	−0.033	0.877	.97	0.967	0.174–5.395
Organ involvement	−20.672	6485.543	.998	1.05E-09	0-Exp (12,689.9)
LMS	−0.056	0.074	.45	0.945	0.817–1.094

Table 5: The diagnostic ability of each paranasal sinus CT finding in differentiation between GPA and EGPA

Imaging Finding	AUROC	95% CI	Sensitivity	Specificity
HU	0.777	0.642–0.88	72.7	90.3
High-density component	0.783	0.648–0.884	72.7	83.9
Bone destruction	0.68	0.538–0.802	90.9	45.2
Sclerotic change	0.644	0.501–0.771	77.3	51.6
Organ involvement	0.597	0.453–0.729	100	19.4

was 62.9 HU), and all the patients who did not have subjective high-density secretions had HUs lower than 60 HU (the highest value was 50.6 HU). The subjective high-density paranasal secretions with HUs higher than 60 have been reported to contain abnormal contents, such as allergic mucin with eosinophilic infiltrates, compared with common secretions in bland cases of rhinosinusitis. The mucus of acute sinus secretions is known to have an attenuation of 10–25 HUs, with the chronic state becoming more viscous and having a higher attenuation of 30–60 HUs.²¹ Furthermore, the mean HU of the highest attenuation portion of the paranasal sinuses and the presence of subjective high-density secretions showed better diagnostic performance in differentiation between GPA and EGPA, compared with bone destruction, sclerotic bone change, and adjacent organ involvement. However, there were also 5 patients with GPA with subjective high-density secretions in the paranasal sinuses. Because patients with GPA sometimes present with epistaxis, the high density may reflect hemorrhagic contents.^{18,19}

Forty-five percent of the patients with GPA and 9.1% of the patients with EGPA had bone destruction in the present study. Two patients with EGPA showed nasal septal perforations as their only manifestation of bony destruction. This result is within the range of previously published studies reporting that approximately 35.9%–75% of patients with GPA showed bone destruction.^{4,5,22–27} It is thought that avascular necrosis resulting from the occlusion of small arteries by an inflammatory cell infiltrate leads to the pathophysiologic process of bone destruction in patients with GPA.^{4,28}

Previous literature has reported that up to 50% of patients with GPA had neo-osteogenesis, resulting in a characteristic sclerotic bony appearance with a leading edge of a well-corticated bone.^{4,5,22,25} The maxillary sinuses are known to be mainly affected.²⁵ Approximately one-half of the patients with GPA (52%) in this study also showed sclerotic bone change (Figs 2C, -D, 3); all of the patients except for 1 had maxillary sinus involvement. In contrast, only 1 patient among the 5 patients showing sclerotic bone changes in the EGPA group had involvement of the maxillary sinuses.

GPA is also well known for involving other head and neck regions along with the nasal cavities and paranasal sinuses, including orbits, nasolacrimal ducts, skull base, ear, and intracranial extension.^{4,5,19,29–31} The orbit has reportedly been involved in 18%–58% of patients with GPA.¹⁹ We also found that 16% of the patients with GPA had orbital or nasolacrimal duct involvement, with 1 patient having intracranial extension (Fig 3D). However, no patient showed additional organ involvement in the EGPA group.

Patients with EGPA usually have sinonasal polyposis, and previous reports have described 52%–76% of these patients as having sinonasal polyps. Patients with EGPA have been described as having nasal polyps more frequently than the GPA group.^{7,17,20,26} Our cohort showed a slightly lower prevalence of nasal polyps in the EGPA patient group (45.5%) than in previous studies. However, it still demonstrated a higher prevalence than the GPA group (32.3%).

The mean LMS of the GPA group and the EGPA group were similar to the results of previous studies.^{15,24,26,32} The patients with EGPA had slightly higher LMS values than the patients with GPA. However, there was no statistically significant difference. A

study with eosinophilic sinus diseases reported that EGPA had a milder form of sinus opacification than other eosinophilic sinus diseases.¹⁵ This might be the reason for the lesser difference in LMS between EGPA and GPA.

There are several limitations in this study. First, the study population was small, and the results of this study may not be generalizable to a large patient population. Further studies on a larger patient cohort are needed to strengthen our results. Second, because most patients did not undergo paranasal sinus surgery, histopathologic confirmation could not be performed for the high-density material in the paranasal sinuses. Only 3 patients in the EGPA group had pathologic results of hypereosinophilic infiltrates from the paranasal sinuses and showed subjective high density on the sinus CT. However, all the subjective high-density sinus opacification had HUs higher than 60, which was beyond the level of usual secretions in chronic sinusitis.

CONCLUSIONS

Patients with EGPA had more subjective high-density paranasal sinus secretions on CT compared with patients with GPA. Additionally, the mean HU of the patients with EGPA was higher than that of the patients with GPA. The HU threshold value to discriminate between GPA and EGPA was 64.3. The presence of subjective high density in the paranasal sinuses or the mean HU of the highest-density portion of the paranasal sinuses independently differentiates EGPA from GPA. In addition to the previously reported CT findings, such as bone destructions, sclerotic bone changes, and adjacent organ involvement, evaluating the high-density opacification in the paranasal sinuses on CT can aid in differentiating sinusitis between the GPA and EGPA.

Disclosure forms provided by the authors are available with the full text and PDF of this article at www.ajnr.org.

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